

Lecture 17: Reduced Row Echelon Form and Row Spaces

1 Row Equivalence

Let F be a field and let $M_{m \times n}(F)$ denote the set of all $m \times n$ matrices over F .

Definition 1 (Elementary Row Operations). The following operations on matrices are called **elementary row operations**:

1. Interchanging two rows.
2. Multiplying a row by a nonzero scalar.
3. Adding a multiple of one row to another row.

Definition 2 (Row Equivalence). Let $A, B \in M_{m \times n}(F)$. We say that A and B are **row equivalent**, written

$$A \sim B$$

if B can be obtained from A by a finite sequence of elementary row operations.

Theorem 1. *Row equivalence is an equivalence relation on $M_{m \times n}(F)$.*

Proof. We verify the three properties.

Reflexivity: No row operations are required to transform A into itself.

Symmetry: Each elementary row operation has an inverse.

Transitivity: A composition of sequences of row operations is again a sequence of row operations. $\square$

2 Reduced Row Echelon Form

Definition 3 (Reduced Row Echelon Form). A matrix $A \in M_{m \times n}(F)$ is said to be in **reduced row echelon form (RREF)** if it satisfies:

1. All nonzero rows appear above any zero rows.
2. The first nonzero entry in each nonzero row is 1. This entry is called a **leading 1**.
3. Each leading 1 is the only nonzero entry in its column.
4. The leading 1 in each row occurs strictly to the right of the leading 1 in the row above.

Example 1. The matrix

$$\begin{pmatrix} 1 & 2 & 0 & 0 & -3 & 0 \\ 0 & 0 & 1 & 0 & 7 & 0 \\ 0 & 0 & 0 & 1 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

is in reduced row echelon form.

However,

$$\begin{pmatrix} 1 & 2 & 1 & 0 & -3 & 0 \\ 0 & 0 & 1 & 0 & 7 & 0 \end{pmatrix}$$

is not in RREF because the pivot column contains multiple nonzero entries.

3 Existence of RREF

Theorem 2. Every matrix $A \in M_{m \times n}(F)$ is row equivalent to a matrix in reduced row echelon form.

Proof. We perform the following algorithm.

Let j_1 be the smallest column index such that column j_1 contains a nonzero entry.

Step 1. Use row swaps to move a nonzero entry into position $(1, j_1)$.

Multiply the first row by a scalar to make this entry equal to 1.

Use row operations to eliminate all other entries in column j_1 .

Step 2. Let $j_2 > j_1$ be the smallest index such that some entry in rows $2, \dots, m$ of column j_2 is nonzero.

Swap rows if necessary to move a nonzero entry to position $(2, j_2)$.

Scale the row so the entry becomes 1.

Eliminate all other entries in column j_2 .

Step 3. Continue this process until no further pivots can be found.

The resulting matrix satisfies all four conditions of RREF. □

4 Row Space

Definition 4. Let $A \in M_{m \times n}(F)$.

The **row space** of A is the subspace of F^n spanned by the rows of A .

Lemma 1. Elementary row operations do not change the row space of a matrix.

Proof. Each row operation replaces a row by a linear combination of rows.

Therefore the span of the rows remains unchanged. □

5 Uniqueness of RREF

Theorem 3. Two matrices in reduced row echelon form with the same row space are equal.

Proof. Let

$$A = \begin{pmatrix} R_1 \\ \vdots \\ R_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} S_1 \\ \vdots \\ S_u \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

be matrices in RREF.

Assume

$$\text{row space}(A) = \text{row space}(B).$$

Step 1: $R_1, \dots, R_t$ form a basis of the row space of A .

Because each row contains a pivot position that is zero in all other rows, any linear dependence would force all coefficients to be zero.

Thus the rows are linearly independent.

Similarly, $S_1, \dots, S_u$ form a basis of the row space of B .

Since the row spaces are equal,

$$t = u.$$

Step 2: Because the pivot columns must coincide, the pivot positions in both matrices are identical.

Thus each corresponding row must be identical:

$$R_i = S_i.$$

Therefore $A = B$. □

6 Consequences

Corollary 1. *Every matrix is row equivalent to a unique matrix in reduced row echelon form.*

Proof. Existence follows from Gaussian elimination.

Uniqueness follows from the previous theorem. □

Corollary 2. *Two matrices have the same row space if and only if they are row equivalent.*

Proof. If $A \sim B$, then elementary row operations preserve row space.

Conversely, if the row spaces are equal, their RREF forms must be identical.

Hence A and B are row equivalent. □